

POWER_BI_BUILD_GUIDE

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Power BI Build Guide

Real Estate AI Decision Intelligence Platform — Phase 10A

This guide gets you from the 12 data-mart CSVs to a finished, presentable `.pbix` file. Follow it top to bottom. Everything referenced here is in `phase10a/data_marts/` plus 3 extra files in this kit (`mart_01b_decision_distribution.csv`, `power_bi_theme.json`, `DAX_measures.md`).

Estimated time: 3-4 hours for all 7 pages.

STEP 0 — Files you need in one folder before starting

Copy these into a single folder (e.g. `powerbi_build/`): - All 12 files from `phase10a/data_marts/*.csv` - `mart_01b_decision_distribution.csv` (from this kit) - `power_bi_theme.json` (from this kit) - Keep `DAX_measures.md` open in a text editor for copy-pasting as you go

STEP 1 — Import all 13 CSVs

1. Open Power BI Desktop → **Home** → **Get Data** → **Text/CSV**
 2. Import each file one at a time (or use **Get Data** → **Folder** and point at your `powerbi_build/` folder to bring in all 13 at once)
 3. For each, click **Transform Data** and check:
 - `property_id` / `locality_id` columns are typed as **Whole Number**
 - Percentage columns (`irr`, `roi_total_holding_period`, `probability_negative_npv`, `yoy_price_growth_pct`, etc.) are typed as **Decimal Number** — you will format them as % at the visual level, don't multiply by 100 in Power Query
 - `decision`, `status`, `scenario`, `source`, `topic` columns are typed as **Text**
 4. Click **Close & Apply**
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STEP 2 — Apply the theme

View → **Themes** → **Browse for themes** → select `power_bi_theme.json`. This gives you the same dark palette as the HTML mockups (gold accent `#D9A85C`, green `#4FAE84` for positive, red `#D9695C` for negative).

STEP 3 — Build relationships (Model view)

Click **Model** on the left rail. Drag to create these relationships:

From	To	Key	Cardinality
<code>mart_03_property_investment</code>	<code>mart_04_financial_risk</code>	<code>property_id</code>	1:1
<code>mart_03_property_investment</code>	<code>mart_06_decision_summary</code>	<code>property_id</code>	1:1
<code>mart_03_property_investment</code>	<code>mart_04b_scenarios_base3</code>	<code>property_id</code>	1:many
<code>mart_03_property_investment</code>	<code>mart_04c_scenarios_extended4</code>	<code>property_id</code>	1:many
<code>mart_06_decision_summary</code>	<code>mart_06b_decision_components</code>	<code>property_id</code>	1:many
<code>mart_06_decision_summary</code>	<code>mart_06c_decision_gates</code>	<code>property_id</code>	1:many
<code>mart_06_decision_summary</code>	<code>mart_06d_decision_reasons</code>	<code>property_id</code>	1:many
<code>mart_02_market_locality</code>	<code>mart_05_geo</code>	<code>locality_id</code>	1:1

Leave unconnected (by design, not an oversight): - `mart_01_portfolio_kpis` — single summary row, used standalone on Page 1 - `mart_07_data_quality` — standalone qualitative table, used on Page 7 - Property-level marts (03/04/06...) have no `locality_id` column (only `locality_name` text) — so they are NOT related to `mart_02` / `mart_05` in this build. If you want cross-filtering between the property pages and the locality pages later, relate on `locality_name` (text relationship) — but note two different cities could theoretically share a locality name, so treat this as an optional advanced step, not a default.

STEP 4 — Add the DAX measures

Open `DAX_measures.md` and paste each measure into its listed table (right-click the table in the **Fields** pane → **New measure**). Do this now so they're available while building visuals below.

STEP 5 — Build the 7 report pages

Add 7 pages (right-click the page tab → rename each). For every page, first add a **Text box** at the top with the page title, matching the HTML mockup headers.

Page 1 — Executive Overview

Source: `mart_01_portfolio_kpis`, `mart_01b_decision_distribution`

Visual	Type	Fields
Top KPI row (4 cards)	Card	<code>total_properties_in_platform</code> , <code>total_localities</code> , <code>mean_decision_score</code> , <code>mean_decision_confidence</code>
Decision split	Donut chart	Legend = <code>decision</code> , Values = count (from <code>mart_01b_decision_distribution</code>)
Coverage KPI	Card + gauge	[<code>Pct Fully Evaluated</code>] measure
Invest rate	Card	[<code>Pct Invest Rate</code>] measure

Conditional formatting: on the donut chart, set data colors manually — INVEST= `#4FAE84`, HOLD= `#D9A85C`, AVOID= `#D9695C`, INSUFFICIENT= `#5D6B7C` (right-click the visual → Format → Colors → click each legend swatch).

Page 2 — Market & Locality Intelligence

Source: `mart_02_market_locality`

Visual	Type	Fields
Price growth by locality	Clustered bar chart	Axis = <code>locality_name</code> , Values = <code>yoy_price_growth_pct</code> , sort descending
Demand vs supply	Scatter chart	X = <code>latest_demand_index</code> , Y = <code>latest_supply_index</code> , Size = <code>location_score</code> , Details = <code>locality_name</code>
Top localities table	Table	<code>locality_name</code> , <code>city_name</code> , <code>latest_price_sqft</code> , <code>growth_rank</code> , <code>demand_supply_rank</code> (sort by <code>growth_rank</code> ascending)
Slicer	Slicer	<code>city_name</code>

Apply `[Growth Color]` measure to the bar chart via Conditional Formatting → Font color → Format by: Field value → `Growth Color`.

Page 3 — Property & Investment Analysis

Source: `mart_03_property_investment`

Visual	Type	Fields
Property table	Table/Matrix	<code>property_id</code> , <code>property_type</code> , <code>area_sqft</code> , <code>asking_price</code> , <code>predicted_valuation</code> , <code>valuation_gap_pct</code> , <code>decision</code>
Valuation gap chart	Bar chart	Axis = <code>property_id</code> , Values = <code>valuation_gap_pct</code>
Slicer	Slicer	<code>decision</code>
Property count	Card	<code>[Property Count]</code> measure

Apply `[Valuation Gap Color]` to the bar chart (same conditional-formatting method as Page 2).

Page 4 — Financial & Risk Intelligence

Source: `mart_04_financial_risk`, `mart_04c_scenarios_extended4`

Visual	Type	Fields
Financial summary table	Table	<code>property_id</code> , <code>irr</code> , <code>npv</code> , <code>cap_rate</code> , <code>dscr</code> , <code>roi_total_holding_period</code>
Scenario spread	Line/column chart	Axis = <code>scenario</code> (sorted by the <code>Scenario Order</code> column you added), Values = <code>irr</code> , Legend = <code>property_id</code>
Risk KPIs row	Cards	<code>[Prob Negative NPV Pct]</code> , <code>probability_meets_target_irr</code> , <code>var_95_npv</code> , <code>cvar_95_npv</code>
Cap rate vs cash-on-cash	Scatter	X = <code>cap_rate</code> , Y = <code>cash_on_cash_return</code> , Details = <code>property_id</code>

Add a text box on this page: “*UNCALIBRATED — every probability/VaR/ CVaR figure comes from Phase 7’s Monte Carlo engine, whose distribution bounds are documented judgement calls, not fitted to real historical outcomes.*” This single sentence is what makes this page credible in an interview — don’t skip it.

Page 5 — Geo Intelligence

Source: `mart_05_geo`

Visual	Type	Fields
Locality map	Map (or ArcGIS Map for more control)	Location: <code>latitude / longitude</code> , Size = <code>location_score</code> , Tooltip = <code>locality_name</code> , <code>accessibility_score</code>
Score comparison	Clustered bar	Axis = <code>locality_name</code> , Values = <code>accessibility_score</code> , <code>transit_score</code> , <code>commercial_score</code>
Data-gap callout	Card	[Pct Missing Flood Data] measure, label it “% localities: flood risk data unavailable”

Do not hide or soften the flood-data-gap card — it’s a governance strength, not a weakness, when framed correctly (see the report’s own Section on this).

Page 6 — Decision Intelligence (the most important page)

Source: `mart_06_decision_summary`, `mart_06b_decision_components`, `mart_06c_decision_gates`,
`mart_06d_decision_reasons`

Visual	Type	Fields
Decision matrix	Table/Matrix	<code>property_id</code> , <code>decision</code> , <code>decision_score</code> , <code>decision_confidence</code>
Component breakdown	Stacked bar chart	Axis = <code>property_id</code> , Values = score, Legend = component
Gate pass/fail	Table with icons	<code>property_id</code> , <code>gate</code> , <code>passed</code> (conditional icon: ✓/✗)
Reasons & risks	Table	<code>property_id</code> , <code>type</code> , <code>text</code> (filter by selected property via a slicer or click- to-filter from the matrix)
Gate pass rate	Card	[Gate Pass Rate] measure

Apply [Decision Color] to the decision matrix (Conditional Formatting → Background color → Field value → Decision Color).

Make this page interactive: set the decision matrix as the filter source — click a property row and have the component/gates/reasons visuals cross-filter automatically (Power BI does this by default via the `property_id` relationship). This is the single best “wow” moment for a live demo — click a property, watch its whole reasoning update.

Page 7 — Data Quality & Monitoring

Source: `mart_07_data_quality`

Visual	Type	Fields
Status table	Table	<code>phase</code> , <code>metric</code> , <code>value</code> , <code>status</code>
Rejected models callout	Card	[Rejected Model Count] measure
Status distribution	Bar chart	Axis = <code>status</code> , Values = count of rows

Apply [Status Color] to the status table’s `status` column (Conditional Formatting → Font color → Field value).

Talking point for this page in an interview: point at the REJECTED row and explain *why* (e.g. days-on-market model showed $R^2 \approx 0$, sale probability showed ROC-AUC ≈ 0.52 — both confirmed near-random and withheld from production use). This is what separates this project from most portfolio ML projects.

STEP 6 — Add page navigation

Insert → Buttons → Blank button on each page, styled as a small nav bar (matches the HTML mockup's top-right nav). Set each button's action to **Page navigation** → target page. Copy one button, paste 7 times, retarget each.

STEP 7 — Save and rehearse

1. **File → Save As** → `RealEstateAI_DecisionIntelligence.pbix`
 2. Before presenting: click through all 7 pages once, click a property on Page 6 to confirm cross-filtering works, and re-read the Page 4 UNCALIBRATED text box out loud once — it should feel natural to say in an interview, not like a disclaimer you're hiding.
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Common Import Issues

- **Percentages showing as 1785% instead of 17.85%:** you formatted the visual as a percentage AND the source is already $\times 100$. Check whether the mart column is a raw decimal (0.1785) or already a percent number — `irr`, `roi_total_holding_period`, `probability_*` are raw decimals in every mart; format as % at the visual level, don't transform the data.
- **Relationships showing as dotted/inactive lines:** Power BI only allows one active relationship per table pair by default — if you see a dotted line, right-click it → Properties → make it active, or use `USERELATIONSHIP()` in a measure if you deliberately need both.
- **Map visual shows nothing:** confirm `latitude/longitude` are typed as Decimal Number, not Text, in Power Query.